

Closed-Loop LLM Agents for Multi-UAV Wildfire Rescue

Author One

Department, Institution, City, Postal Code, Country

Author Two

Department, Institution, City, Postal Code, Country

August 17, 2026

Abstract

Multi-UAV wildfire rescue requires closed-loop decisions that convert uncertain observations into executable fleet actions and adapt to changing conditions. We formulate wildfire rescue as a closed-loop decision problem spanning situation understanding, resource-constrained planning, validation, and runtime adaptation. Our modular large language model (LLM)-based agent framework combines Vision Analysis, Mission Planning, Episodic and Lesson Memory retrieval, independent Validation, and Runtime Replanning to generate and repair multi-UAV missions.

We evaluate the framework in a real-UAV-informed AirSim suite with 80 fixed evaluation episodes across four difficulty levels. Sensing, energy, payload, and communication constraints are calibrated from a physical platform. Experiments show that a conventional optimization baseline matches LLM mission scores with much lower planning time in simple and normal episodes, whereas the LLM configurations achieve higher scores and generally lower replanning latency in hard and expert episodes. Ablations further show that Memory and Runtime Replanning become more important as mission complexity increases. A fine-tuned Qwen3-8B planner retains 98.24% of the Gemini 3.1 Pro teacher planner’s average performance while substantially reducing planning overhead. These results support LLM agents as high-level decision components for adaptive multi-UAV wildfire response. The implementation is available at <https://github.com/tongsuhyem/Pipeline>.

Keywords: multi-UAV wildfire response; LLM agents; compact LLM fine-tuning; closed-loop decision making; real-to-sim evaluation

1 Introduction

Wildfires threaten ecosystems, infrastructure, and human safety (?). UAVs can rapidly observe hazardous and inaccessible areas through flexible aerial sensing (?), but operational response requires more than aerial detection. Observations must be converted into task zones, assigned across a resource-limited fleet, and revised as the fire or UAV state changes. Multi-UAV wildfire rescue is therefore a closed-loop decision problem rather than an isolated perception or path-planning task.

Existing work has advanced fire detection, aerial scene understanding, and damage assessment, but transforming observations into coordinated response actions remains challenging (?). A practical plan must respect energy, payload, communication, safety, and priority constraints (?), while accommodating fire evolution, UAV faults, and resource shortages. Large language models (LLMs) and multimodal LLMs offer a flexible interface for connecting visual interpretation, task decomposition, and high-level planning (?). However, free-form generation is insufficient for safety-critical missions: plans can violate fleet constraints or become invalid when execution conditions change.

We formulate multi-UAV wildfire rescue as a decision chain from aerial observation to fire-situation understanding, resource-constrained planning, and runtime adaptation. Our modular system combines Vision Analysis, Mission Planning, Semantic and Deterministic Validation, Episodic and Lesson Memory, and Runtime Replanning. It converts aerial imagery into fire points and task zones, generates an executable plan, and uses retrieved cases and lessons to support initial planning and fault-driven repair.

To evaluate this capability, we construct a real-UAV-informed AirSim suite that assesses understanding, constrained planning, and adaptation under runtime perturbations. It calibrates observation, energy, payload, and communication conditions from a real platform, organizes 80 fixed evaluation episodes into four difficulty levels, and measures completion, response, mission time, event stability, resource pressure, and coordination.

The main contributions of this work are summarized as follows:

1. We develop a modular LLM-based rescue system that integrates Vision Analysis, constrained multi-UAV Mission Planning, independent Validation, Episodic and Lesson Memory, and Runtime Replanning.
2. We construct a real-UAV-informed AirSim evaluation suite with four difficulty levels and six mission-level metrics for end-to-end assessment under controlled operational pressure.
3. We compare four LLM-based configurations with an Optimization-Based + Human baseline, analyze the contributions of Validation, Memory, and Runtime Replanning, and assess whether a compact planner specialized with validated demonstrations can retain strong mission performance with lower planning overhead.

2 Related Work

2.1 UAV-based Wildfire Response

UAVs provide a practical means of observing wildfires in hazardous and difficult-to-access areas, but operational wildfire response requires more than detecting fire in aerial imagery. It requires a decision chain that transforms observations into a current representation of the incident, allocates limited airborne resources to the resulting tasks, and updates the mission when the environment or fleet state changes. Existing work has made substantial progress on the individual stages of this chain, including aerial perception, active situation monitoring, autonomous flight planning, and multi-UAV coordination.

At the perception stage, multimodal sensing improves the reliability of fire observations under the visually challenging conditions common in wildfire scenes. Thermal- RGB fusion has been used to detect fire and hotspots from UAV imagery, reducing sensitivity to smoke and illumination variation (?). Transformer-based aerial semantic segmentation further supports the interpretation of burned regions, vegetation, and terrain (?), while

learning-based 3D reconstruction can provide geometric environmental models for navigation and inspection (?). These methods produce the visual and spatial evidence needed for response, but their outputs must still be converted into mission-level entities, such as fire points, task zones, and safety-relevant context, before they can drive rescue actions.

Situation awareness and planning research addresses part of this conversion. Autonomous exploration and mapping enable UAVs to update knowledge of unknown or changing disaster environments (?); next-best-view planning selects viewpoints that maximize information gain and reduces observation uncertainty (?). On the action side, deep reinforcement learning has been applied to autonomous navigation in unknown environments (?), and coverage path planning supports large-area search and perimeter scanning (?). However, these methods generally optimize sensing, navigation, or coverage as an individual capability. They do not by themselves produce a fleet-level, resource-constrained response plan that explicitly links evolving fire information to task priorities, payload use, endurance, and execution status.

Multi-UAV methods extend decision making from a single vehicle to a team. Multi-agent reinforcement learning has demonstrated cooperative navigation and task allocation capabilities (?), while the broader disaster-response literature highlights the value of UAV support and human-robot collaboration in search-and-rescue operations (?). Nevertheless, wildfire response remains difficult because fire evolution, communication changes, energy depletion, and vehicle failures can invalidate an initially feasible allocation. The literature is therefore still dominated by modular or open-loop solutions: perception, mapping, path planning, and coordination are often evaluated separately. In contrast, this work treats wildfire rescue as a closed-loop decision problem from aerial observation to structured fire-situation understanding, resource-constrained multi-UAV planning, validation, and local runtime replanning. This formulation connects the outputs of perception to executable team actions and preserves mission continuity under operational perturbations.

2.2 LLM-Based Robotic Agents

Large language models (LLMs) provide an increasingly capable interface between high-level human intent and robot behavior. Rather than replacing low-level perception or control modules, they are commonly used to interpret language, decompose goals, select skills, and reason over structured observations. This role is particularly relevant to wildfire response, where a decision system must translate a changing fire situation and operational constraints into a coherent fleet mission rather than execute a fixed motion policy.

Language grounding and hierarchical planning are central directions in this area. Say-Can combines language-model scoring with robot affordance estimates so that high-level instructions are grounded in executable skills (?). Code as Policies uses language models to generate programmatic control policies, providing a flexible mechanism for composing robot behavior from functions and APIs (?). Related design guidance has shown how ChatGPT can be used to generate robot workflows while exposing tool descriptions and environmental constraints to the model (?). Together, these approaches show that LLMs can support task decomposition and action sequencing, but they also make clear that generated actions require grounding and external feasibility checks.

Embodied and multimodal models extend this reasoning interface to visual and physical state. RT-1 learns vision-language-action policies from large-scale real-robot data (?), while PaLM-E incorporates visual inputs and robot states into an embodied language model (?). Socratic Models further demonstrate zero-shot reasoning by composing

specialized vision and language models (?). These methods strengthen the connection between perception and action, yet they are generally evaluated on individual robots and predefined manipulation or navigation tasks. They do not directly address the resource-constrained allocation of dynamic, interdependent tasks across a rescue fleet.

LLM-based agents have also explored closed-loop decision making through environmental feedback, navigation, and experience reuse. Inner Monologue uses language-based feedback to revise embodied plans (?), and LM-Nav combines large pretrained models with semantic waypoints for open-world navigation (?). Voyager couples an LLM agent with an automatic curriculum and a growing skill library for open-ended exploration (?). Nevertheless, these systems primarily target single-agent embodied environments. In contrast, our system applies LLM reasoning to multi-UAV wildfire response, coupling multimodal fire-situation understanding with constrained fleet-level planning, independent Validation, Episodic and Lesson Memory, and Runtime Replanning. The resulting design treats the LLM as a high-level decision component within a closed-loop safety and execution framework rather than as an unconstrained controller.

3 Task Modeling and System Design

3.1 Task Modeling

Wildfire rescue is a closed-loop decision task rather than an isolated perception or allocation problem. In an operational mission, aerial observations first reveal dispersed fire points; these points must then be organized into actionable task zones. The fleet must allocate its limited energy, payload, and flight time to these regions, while unexpected failures or changes in the fire situation may require the current plan to be revised during execution. We therefore abstract the real rescue workflow as a sequence from situation understanding to resource-constrained planning and runtime adaptation.

For each mission, the rescue world is represented by the structured state

$$S = \{O, F, R, U, C, E\}.$$

Here, O denotes aerial observations; F denotes fire points; R denotes task zones; U denotes UAV operational states; C denotes mission constraints; and E denotes runtime perturbations. This representation separates the underlying rescue world from the information available to the decision system. At mission start, the system receives $X_0 = \{O, U, C\}$, whereas F and R must be inferred from the observation. When a runtime perturbation occurs, it receives the updated operational state U_t and the observed perturbation E_t . The required outputs are an initial mission plan P and, when necessary, a revised plan P' .

The resulting task model comprises the following decision chain:

$$\begin{aligned} O \rightarrow F \rightarrow R, \quad (R, U, C) \rightarrow P, \\ (P, U_t, E_t) \rightarrow P'. \end{aligned}$$

The first two relations transform raw aerial observations into an actionable regional representation of the fire situation. The third relation generates an executable fleet plan under operational constraints. The final relation adapts that plan to changes encountered during execution. This formulation captures the continuous rescue workflow while retaining the stage-specific information needed to design and evaluate a closed-loop system.

3.2 System Overview

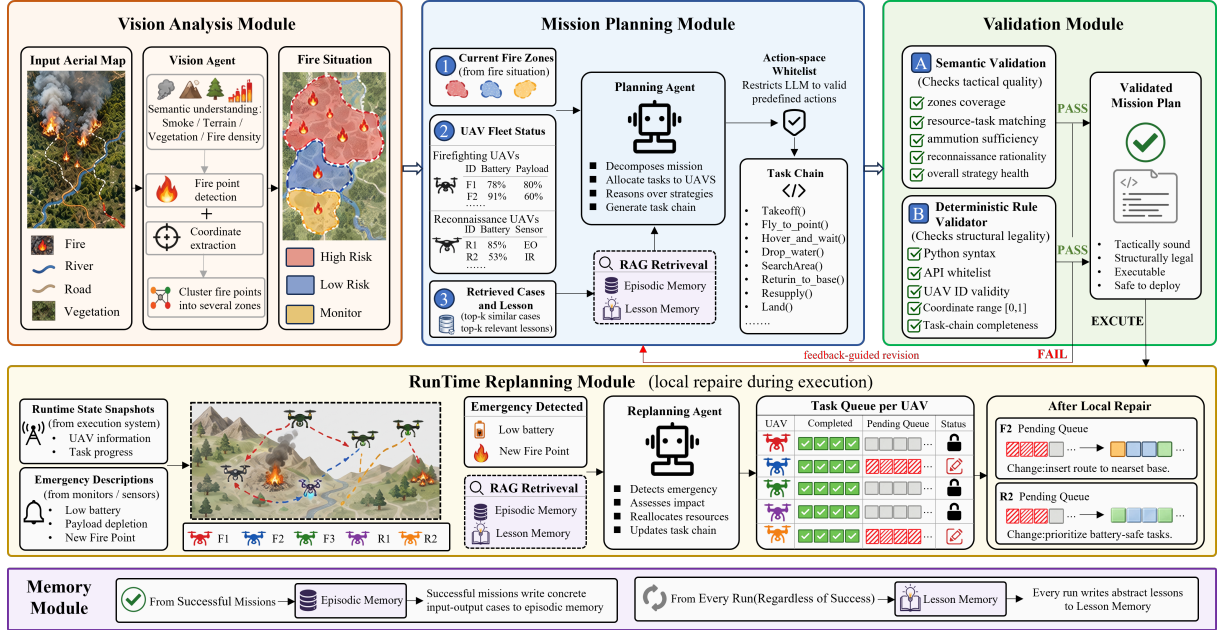


Figure 1: Overview of the proposed system.

Based on the task model above, the proposed system separates initial mission generation, independent plan checking, runtime adaptation, and experience reuse. The system takes aerial fire imagery as the initial input and produces an executable mission plan for a multi-UAV fleet. The main planning pipeline contains three modules: Vision Analysis, Mission Planning, and Validation. A Runtime Replanning module operates outside the pipeline and adjusts the plan when unexpected events occur during execution. Two memory modules, Episodic Memory and Lesson Memory, supply past experience to the planning and replanning modules. Figure 1 shows the resulting architecture. Vision Analysis implements the transformation from observations to fire points and task zones; Mission Planning maps task zones, fleet states, and constraints to an initial plan; and Runtime Replanning produces local revisions after runtime perturbations.

Vision Analysis. Vision Analysis converts raw aerial imagery into a structured description of the fire situation. This module uses a multimodal LLM to perform two tasks in a single inference pass. First, it detects all visible fire points in the image and outputs their normalized coordinates. Second, it clusters these points by spatial density, partitioning the continuous fire field into several discrete task zones.

This module uses a multimodal LLM rather than a vision-only model because zone division depends on more than fire point locations. The division also depends on visual context, such as smoke, vegetation, and terrain boundaries. These cues require joint reasoning over appearance and spatial semantics.

Mission Planning. Mission Planning is the core decision module. It converts the structured fire situation produced by Vision Analysis into an executable multi-UAV collaborative plan. A planning LLM generates a complete task chain for each UAV in the fleet, covering task allocation, temporal sequencing, and endurance management. To keep outputs controllable, the LLM is restricted to a predefined set of allowable actions.

Mission Planning combines the fire situation, retrieved experience, and operational constraints. It receives the structured fire situation from Vision Analysis, retrieves relevant past cases and lessons from both memory modules, and produces a coordinated plan in which each UAV’s task chain respects the fleet’s capabilities and the mission’s operational constraints. Formally, the planning process is expressed as

$$P = \mathcal{G}(R, U, C, M_E(q), M_L(q)), \quad q = q(R, U, C), \quad (1)$$

where $\mathcal{G}$ denotes the planning LLM, q is a query constructed from the current task zones, fleet states, and constraints, and $M_E(q)$ and $M_L(q)$ denote the episodic cases and lessons retrieved for that query, respectively.

Validation. The Validation module assesses each candidate plan through two sequential checks. Together, the two checks form the audit–correction loop: when either one rejects a plan, its feedback is routed back to Mission Planning for revision. This loop turns validation failures into targeted feedback for the next planning attempt.

The first check, Semantic Validation, uses an independent LLM to evaluate the plan from a tactical rationality perspective. It examines zone coverage completeness, resource–task matching, payload adequacy, and overall strategic health. Using a separate LLM rather than asking the planning LLM to self-check avoids self-consistency bias.

The second check, Deterministic Rule Validation, uses a rule engine to verify structural legality, including UAV existence, coordinate validity, and task chain completeness. Unlike the semantic check, this rule check asks whether the plan is structurally legal. This question can be answered deterministically within the defined rule set.

Runtime Replanning. This module handles runtime perturbations during mission execution. It takes a runtime state snapshot as input, including current positions, battery levels, payloads, and task progress for each UAV. Its output is not a new global plan, but a local revision of the current plan. Unlike the main pipeline, Runtime Replanning reasons about a dynamic execution state and changes only the affected part of the plan.

When a runtime perturbation occurs, the module retrieves similar fault cases from Episodic Memory as repair references. These cases describe how comparable failures were resolved in past missions, providing the replanning LLM with tactical precedents for the current situation.

Memory System. The Memory System allows the system to reuse past experience. The two memory modules differ in what they store and how their contents are used.

Episodic Memory stores concrete successful cases as input–output pairs. Each case records the situation, the generated or revised plan, and the corresponding outcome. Cases are indexed by situational features, so that when a new mission situation arises, the system retrieves the most similar past cases as tactical references. Specifically, the episodic memory is represented as

$$M_E = \{(z_i, P_i, o_i)\}_{i=1}^{N_E}, \quad (2)$$

where z_i , P_i , and o_i denote the task situation, the associated plan, and the execution outcome of case i , respectively. For either episodic memory or lesson memory, the system retrieves the K most semantically similar entries to the current query according to

$$\mathcal{R}_K(q, M) = \underset{m_i \in M}{\text{TopK}} \text{ sim}(q, m_i), \quad \text{sim}(q, m_i) = \frac{\phi(q)^\top \phi(m_i)}{\|\phi(q)\|_2 \|\phi(m_i)\|_2}, \quad (3)$$

where $M \in \{M_E, M_L\}$ is the queried memory, $\phi(\cdot)$ maps a query or memory entry to its semantic embedding, and $\text{sim}(\cdot, \cdot)$ is the cosine similarity. The retrieved set is supplied to the corresponding planning or replanning prompt as reference information.

Lesson Memory serves a different role. Instead of storing concrete cases, it keeps higher-level lessons distilled from every run. After each mission, an LLM reviews the full execution trace. The trace includes the initial plan, validation feedback, runtime perturbations, replanning actions, and final scores. The LLM then extracts concise, reusable lessons. These lessons capture patterns such as common planning mistakes, resource estimation biases, and effective repair strategies.

4 Evaluation Suite Design

Existing datasets and evaluation settings only partially cover the closed-loop rescue decision chain defined in Section 3. As summarized in Table 1, no existing setting jointly evaluates the transition from aerial observation to task-zone construction, resource-constrained fleet planning, and runtime replanning.

Table 1: Coverage of evaluation capabilities in existing datasets, benchmarks, and the evaluation suite used in this work. • indicates explicit coverage, ◦ indicates partial coverage, and — indicates no explicit coverage.

Benchmark / Dataset	Forest Fire Perception	Disaster Semantics	UAV-level Constraints	Runtime Replanning
DOTA (?)	•	—	—	—
VisDrone (?)	•	—	—	—
UAVs-FFDB (?)	•	•	—	—
FLAME (?)	•	•	—	—
MS-FSDB (?)	◦	•	—	—
WildFireVQA (?)	•	•	◦	—
RMA SBench (?)	—	•	◦	◦
CrisisBench (?)	—	•	—	—
Our Evaluation Suite	•	•	•	•

To close this gap, we instantiate the task model of Section 3 as a controlled, real-UAV-informed evaluation suite in AirSim. The suite evaluates whether a system can complete the full rescue loop within a single episode, from aerial fire-situation understanding to an executable plan and its adaptation after runtime perturbations. Its design combines three complementary elements.

First, real-to-sim calibration constrains simulated platforms, sensing, and mission conditions by the capabilities and safety limits of a real UAV. The resulting evaluation episodes are therefore physically and operationally grounded rather than arbitrarily configured. Second, difficulty stratification varies the complexity of fire-point layouts, fleet-resource conditions, and runtime perturbations while preserving a common task definition. It enables controlled comparison of models as closed-loop demands increase. Third, mission-level metrics assess the outcome of the complete rescue process from complementary operational perspectives, rather than collapsing performance into a single score. The following sections detail these three elements.

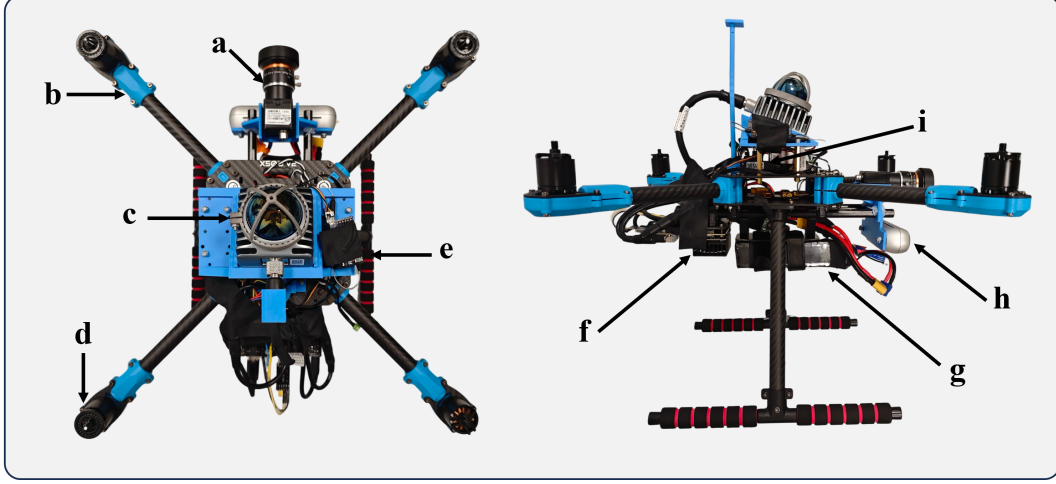


Figure 2: UAV platform and its main components:(a) LiDAR, (b) frame, (c) monocular camera, (d) motor, (e) receiver, (f) onboard computer, (g) power supply, (h) stereo camera, and (i) flight controller.

4.1 Real-to-Sim Calibration

The first design property is realized through a co-design between the real UAV platform and the AirSim simulation environment. As stated above, real UAV parameters constrain every evaluation episode. Accordingly, the real platform defines the engineering boundary of the rescue mission, while AirSim configures controllable and repeatable evaluation episodes within that boundary. Rather than treating the two as separate experimental settings, we use the real platform to define the feasible set from which AirSim configures each simulated UAV and its environment. For each UAV, the configured mission must satisfy

$$\mathcal{E}_{\text{UAV}} = \left\{ (h, v, m, e_{\text{avail}}, d) \left| \begin{array}{l} h \in \mathcal{H}_{\text{safe}}, \quad v \in \mathcal{V}_{\text{safe}}, \\ m \leq m_{\text{max}}, \quad e_{\text{req}} + e_{\text{res}} \leq e_{\text{avail}}, \\ d \leq d_{\text{link}} \end{array} \right. \right\}, \quad (4)$$

where h , v , m , e_{avail} , and d denote flight altitude, velocity, payload mass, available energy, and distance to the coordination link. The safe altitude and velocity ranges are specified by the operating policy. The payload and energy bounds are grounded in the airframe, propulsion system, and battery; the energy reserve enforces a safe return; and the link bound is determined by the deployed communication configuration. Thus, the numerical bounds combine hardware specifications, flight-safety policy, and deployment configuration. Any configuration that violates Eq. 4 is excluded before episode generation. The real UAV platform is shown in Figure 2, and its hardware specifications are listed in Table 2.

The virtual observation process is further fixed by the real camera geometry. Under a nadir-view, locally planar-ground approximation, for altitude h , horizontal field of view θ_{FOV} , and horizontal resolution N_x , the image footprint and ground sampling distance are

$$W(h) = 2h \tan \left(\frac{\theta_{\text{FOV}}}{2} \right), \quad \text{GSD}(h) = \frac{W(h)}{N_x}. \quad (5)$$

Thus, the AirSim camera pose and the scale, density, and placement of fire points are jointly constrained by what the real camera could observe, rather than selected independently. Energy and payload limits likewise constrain route length, task duration, and

Table 2: Technical specifications of the real UAV platform.

Subsystem	Model	Key specifications
UAV frame	Holybro X500 V2	Carbon-fiber quadrotor frame, 500-mm wheelbase, approximately 610-g frame weight, and 1.5-kg published payload capacity excluding the battery at 70% throttle.
Flight controller	Holybro Pixhawk 6X	STM32H753 processor at 480 MHz, triple-redundant IMUs, dual barometers, 16 PWM outputs, two CAN buses, and 100-Mbps Ethernet.
Companion computer	NVIDIA Jetson Orin NX 8 GB	6-core Arm Cortex-A78AE CPU, NVIDIA Ampere GPU, 8-GB LPDDR5 memory, up to 117 TOPS, and configurable 10–40-W power modes.
LiDAR	Livox Mid-360	360° horizontal and -7° – 52° vertical field of view, 200,000 points/s, 10-Hz frame rate, 40-m range at 10% reflectivity, 0.1-m blind zone, 265-g weight, and 6.5-W average power.
RGB-D camera	Intel RealSense D455	Global-shutter stereo depth camera, $87^\circ \times 58^\circ$ depth field of view, depth resolution up to 1280×720 at 90 fps, approximately 0.6–6-m effective range, integrated IMU, USB 3.1 interface, and 116-g weight.
Propulsion system	4× Holybro 2216 KV920	Four 920-KV brushless motors, four 20-A ESCs, and 10×4.5 -in propellers, with a maximum manufacturer bench thrust of approximately 1.33 kgf per motor at 16 V.
Battery	Dualsky XP3300-6HED	6S lithium-polymer battery, 3300-mAh capacity, 22.2-V nominal voltage, 25.2-V fully charged voltage, and approximately 73.3-Wh nominal energy.

suppressible workload. Table 3 records these calibration rules and the resulting simulation settings.

Table 3: Real-to-sim calibration of AirSim UAV and environment settings.

Calibration target	Real-platform basis	AirSim configuration
Imaging geometry	Operating altitude and the RGB-D camera field of view and resolution	Observation height h , camera pose, field of view, and resolution; the resulting footprint $W(h)$ and GSD determine the observable ground area and image scale.
Energy model	Battery capacity, propulsion characteristics, and return-to-base reserve policy	Available energy e_{avail} , reserve e_{res} , and energy consumption for flight and task execution.
Payload model	Airframe and propulsion payload limit	Maximum payload m_{max} and the onboard suppressant inventory for each simulated UAV.
Communication model	Deployed communication configuration and operating range	Link-distance threshold d_{link} and the distance-dependent connectivity condition.

These calibrated bounds provide the common operational basis on which the difficulty levels are constructed.

4.2 Difficulty Stratification

The second design property is that each evaluation episode is purpose-built rather than randomly sampled. Each episode instantiates the task variables defined in Section 3 through three controlled dimensions: fleet and resource conditions, fire-point layouts, and, where applicable, runtime perturbations. Their combinations define four stress-testing difficulty levels in Table 4. Each difficulty level is instantiated by a fixed set of 20 evaluation episodes, yielding 80 episodes in total.

Table 4: Four difficulty levels and their controlled episode conditions.

Difficulty levels	Fire-point configuration	Fleet/resource conditions	Runtime perturbations
Simple	Sparse static fire points; few task zones.	Ample energy and payload.	None.
Normal	Multiple fire points and task zones.	Multi-UAV allocation; normal resource margin.	None.
Hard	Multi-region fire scene.	Energy or payload constrained.	Few, temporally separated perturbations.
Expert	Dense or spreading fire-point layout.	Competing resource demands.	Frequent perturbations; global scheduling pressure.

The fleet and resource conditions in every evaluation episode is constrained by the real-to-sim calibration in Section 4.1. In particular, each simulated UAV must satisfy the feasible operating envelope in Eq. 4, and its imaging, energy, payload, and communication settings follow Table 3. Difficulty is therefore varied only within the engineering and operational limits established for the real platform.

The fire-point configuration of all episode is pre-specified rather than randomly generated. These configuration controls the scene structure and the spatial arrangement and number of fire points per task zone across difficulty levels. Figure 3 provides representative scenes from the four difficulty levels, and Figure 4 summarizes the distribution of fire-point counts per task zone within each difficulty level. Together, they illustrate the scene layouts and task-zone-level fire-point distributions used to instantiate the four difficulty levels.

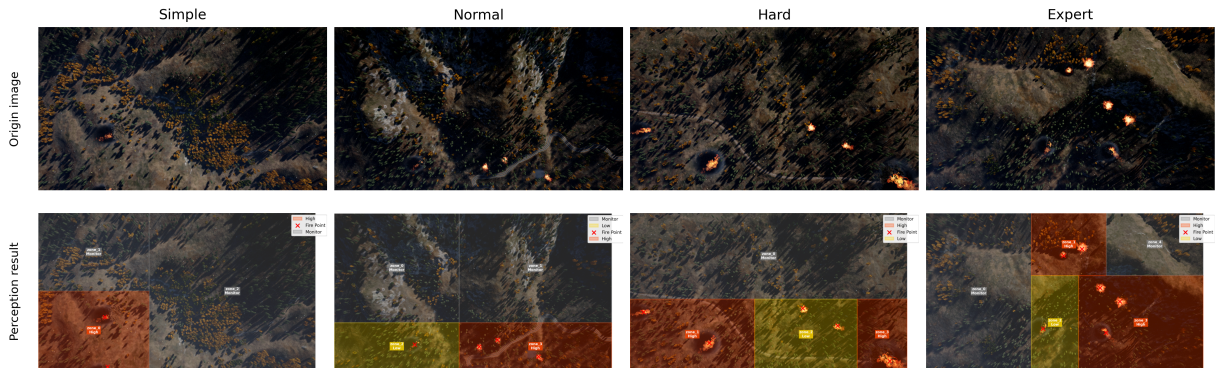


Figure 3: Representative fire scenes from the four difficulty levels. In each example, *Origin image* denotes the original aerial image, and *Perception result* denotes the output generated by the Vision Analysis module, which uses Gemini 3.1 Pro as its foundation model.

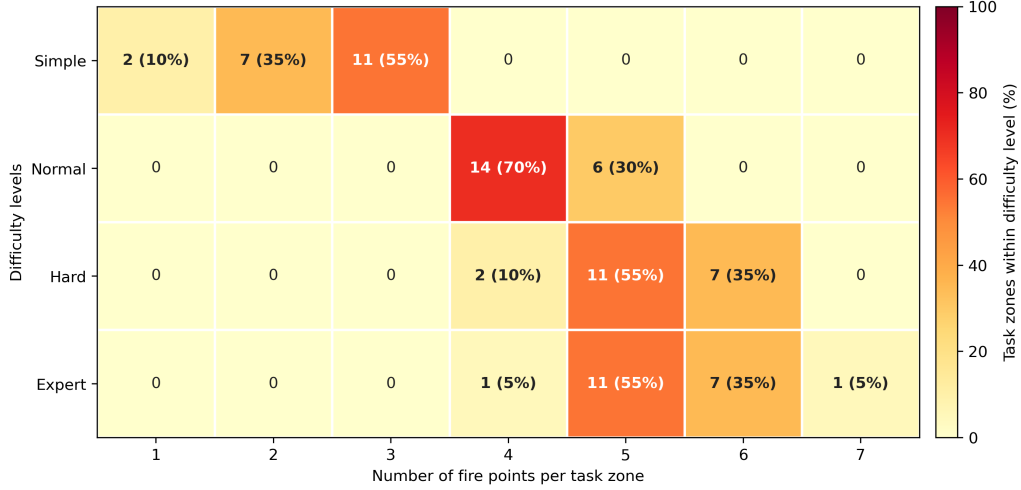


Figure 4: Distribution of fire-point counts per task zone across difficulty levels.

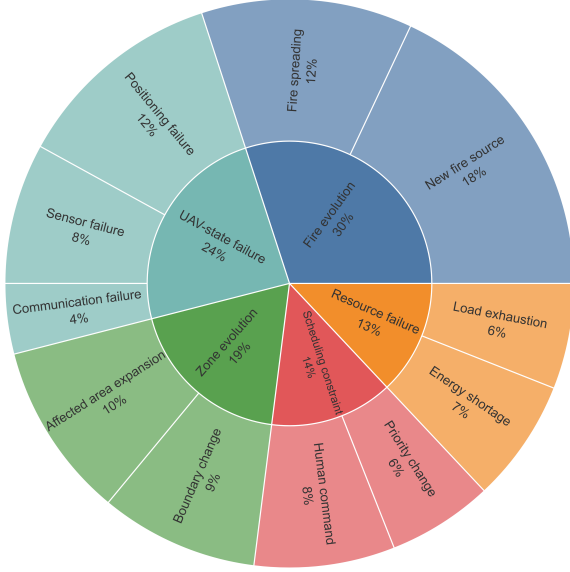
The third controlled dimension introduces runtime perturbations after mission execution has begun. Unlike the preceding two dimensions, which specify the initial fleet state and fire scene, these events modify the operational state or mission constraints faced by an active plan. For this evaluation suite, we construct our own runtime perturbation library comprising five families: fire evolution (30%), UAV-state failure (24%), zone evolution (19%), scheduling constraint (14%), and resource failure (13%). As shown in Figure 5(a), each family contains concrete event types, such as new fire sources and fire spreading, positioning, sensor, and communication failures, boundary changes and affected-area expansion, human commands and priority changes, and load exhaustion and energy shortage.

Runtime perturbations are enabled only in the hard and expert levels and are placed at predefined points in each episode so that replanning behavior can be evaluated under repeatable conditions. During suite construction, events are drawn once from the library according to the configured sampling distribution and then fixed within the evaluation episodes. Hard episodes contain a small number of temporally separated perturbations, whereas expert episodes contain more frequent perturbations that may occur concurrently or in succession and impose stronger communication and global scheduling pressure. Figure 5(b) shows the resulting draws by perturbation family: expert episodes contain more events in every family, while preserving coverage of all five families.

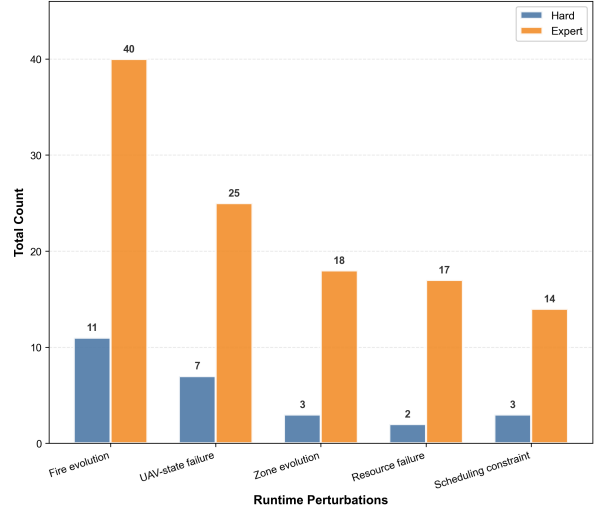
4.3 Evaluation Metrics

We evaluate end-to-end behavior using six complementary mission-level metrics: completion, response, mission time, event stability, resource pressure, and coordination. They measure objective attainment, response timeliness, execution efficiency, continuity under perturbations, resource use, and workload balance, respectively. Event stability is evaluated only in the hard and expert settings, where runtime perturbations are introduced.

All scores are normalized to $[0, 100]$, with higher values indicating better performance. Let Z be the set of task zones, $c_z \in [0, 1]$ the completion ratio of zone z , and ω_z its priority weight. Let $\bar{t}_{\text{resp}}$ and τ denote the priority-weighted mean time to first effective



(a) Runtime perturbation library.



(b) Distribution of perturbations in hard and expert levels.

Figure 5: Runtime perturbations used for closed-loop evaluation.

intervention and the total mission duration. The six scores are defined jointly as

$$\begin{aligned}
 s_{\text{comp}} &= 100 \frac{\sum_{z \in Z} \omega_z c_z}{\sum_{z \in Z} \omega_z}, & s_{\text{resp}} &= 100 \left[1 - \left[\frac{\bar{t}_{\text{resp}} - l_{\text{resp}}}{u_{\text{resp}} - l_{\text{resp}}} \right]_0^1 \right], \\
 s_{\text{time}} &= 100 \left[1 - \left[\frac{\tau - l_{\text{time}}}{u_{\text{time}} - l_{\text{time}}} \right]_0^1 \right], & s_{\text{stab}} &= \frac{100}{N_e} \sum_{e=1}^{N_e} \mathbb{I}(\text{Recovered}(e)), \\
 s_{\text{res}} &= 100 \left[1 - \frac{\bar{p} + \max_j p_j}{2} \right], & s_{\text{coord}} &= 100 \left[1 - \frac{G(\mathbf{n}) + G(\mathbf{d}) + G(\boldsymbol{\tau})}{3} \right].
 \end{aligned} \tag{6}$$

Here, $[x]_0^1 = \min\{1, \max\{0, x\}\}$ clips a value to $[0, 1]$; l_{resp} , u_{resp} , l_{time} , and u_{time} are fixed time-reference thresholds. N_e is the number of runtime perturbations, and $\text{Recovered}(e)$ indicates recovery by a feasible revision within the allowed window. $p_j \in [0, 1]$ and $\bar{p}$ denote the per-UAV and fleet-average resource pressure. $G(\cdot) \in [0, 1]$ is the Gini coefficient of the assigned task counts $\mathbf{n}$, flight distances $\mathbf{d}$, or execution durations $\boldsymbol{\tau}$.

Because no perturbations are introduced in the simple and normal settings, their applicable metric set $\mathcal{K}_d$ contains completion, response, mission time, resource pressure, and coordination; for the hard and expert settings, it additionally contains event stability. For each episode i at difficulty level d , the score combines the weighted aggregation of its applicable metrics with a bounded difficulty reward. The reported score for level d is the mean across its N_d episodes:

$$S_d = \frac{1}{N_d} \sum_{i=1}^{N_d} \left[\sum_{k \in \mathcal{K}_d} w_{k,d} s_{i,k} \frac{1 + \lambda \rho_i}{1 + \lambda} \right], \quad w_{k,d} \geq 0, \quad \sum_{k \in \mathcal{K}_d} w_{k,d} = 1, \tag{7}$$

where $w_{k,d}$ is the normalized weight of metric k at difficulty level d , $\rho_i \in [0, 1]$ is the difficulty coefficient of episode i , and $\lambda \geq 0$ controls the strength of the difficulty reward. The normalization by $1 + \lambda$ ensures that $S_d \in [0, 100]$. The component scores remain available for diagnostic analysis, while S_d provides a unified measure for overall comparison.

5 Experiments

5.1 Experimental Setup

We evaluate the proposed system using the evaluation suite described in Section 4. The experiments comprise three complementary settings. The first assesses the end-to-end system with different general-purpose foundation models and a conventional planning baseline. The second examines the contribution of the main system components through ablation. The third investigates whether a compact model specialized with domain decision traces can replace the general-purpose model in Mission Planning.

In the first setting, we fix Gemini 3.1 Pro for Vision Analysis and compare four LLM-based configurations—using ChatGPT 5.5, Gemini 3.1 Pro, Claude Opus 4.7, and DeepSeek V4 Pro in the remaining LLM-based modules—with an Optimization-Based + Human baseline under the same visual-analysis output, UAV states, action space, execution environment, episodes, and scoring procedure. The baseline uses an OR-Tools CP-SAT planner in the simple and normal settings; in the hard and expert settings, the same planner generates the initial plan and a human operator revises it after each natural-language runtime perturbation. Each configuration is evaluated on the same 80 episodes, with 20 at each difficulty level. Planning time reports the average time required to produce a valid plan.

For the second setting, we use the DeepSeek V4 Pro configuration and compare the full system with three ablated variants: w/o Validation, w/o Memory, and w/o Runtime Replanning. The first removes the semantic and deterministic audit-correction loop; the second disables retrieval from both Episodic Memory and Lesson Memory; and the third retains the initial plan after runtime perturbations without invoking the repair module. All other models, prompts, and execution settings remain unchanged. All four configurations are evaluated on the same 80 episodes, with 20 episodes at each difficulty level. In the simple and normal settings, where no runtime perturbations occur, the w/o Runtime Replanning variant differs only in that the repair module is disabled and therefore remains inactive throughout execution.

For the third setting, we construct 2,000 mission-planning demonstrations using the proposed system with Gemini 3.1 Pro as the teacher model. The demonstrations are generated from mission situations disjoint from the 80 evaluation episodes. Each demonstration has the form

$$(R, U, C) \rightarrow P,$$

where each demonstration pairs a structured description of the task zones, UAV states, and mission constraints with the final multi-UAV plan accepted after the Semantic Validation and Deterministic Rule Validation loop. We use these validated demonstrations to specialize Qwen3-8B for the planning mapping defined in Section 3. The dataset is divided into 90% for training and 10% for validation. We apply 4-bit quantized low-rank adaptation with a maximum sequence length of 6,144 tokens. The model is trained for three epochs with an effective batch size of 16 and a learning rate of 5×10^{-5} . Fine-tuning is conducted on a single NVIDIA GeForce RTX 4090, with 16 Intel Xeon Gold 6430 vCPUs and 120 GB of system memory.

We evaluate both the original and fine-tuned Qwen3-8B by replacing only the Mission Planning model. Vision Analysis and Semantic Validation continue to use Gemini 3.1 Pro, while Deterministic Rule Validation and all other non-LLM components remain unchanged. The two Qwen3-8B configurations are evaluated on the same 40 simple

and normal episodes used in the first setting, with 20 episodes at each level. Because these levels contain no runtime perturbations, this experiment does not invoke Runtime Replanning or assess event stability. The comparison therefore isolates the effect of planner fine-tuning on the same episode subset.

5.2 Foundation Model and Baseline Comparison

Table 5 compares the Optimization-Based + Human baseline with the four LLM-based configurations across all difficulty levels.

Table 5: Main results for the Optimization-Based + Human baseline and four foundation-model configurations across episode difficulty levels. S is the overall score, and the remaining columns report normalized mission-level scores in $[0, 100]$ (higher is better). Event stability is not applicable to the simple and normal settings and is denoted by “–”. Planning time is reported in seconds per planning decision (lower is better). In the hard and expert settings, the baseline includes human runtime replanning. The best value for each score and planning time at every difficulty level is shown in bold when it is unique.

Difficulty	Configuration	S	Mission-level score						Planning time (s)
			Completion	Response	Time	Event stability	Resource pressure	Coordination	
Simple	Optimization-Based + Human	92.59	100.00	92.34	88.94	–	85.64	82.31	3.42
	ChatGPT 5.5	90.63	100.00	89.18	78.99	–	81.20	79.54	41.42
	Gemini 3.1 Pro	93.23	100.00	88.78	91.06	–	84.45	80.71	33.47
	Claude Opus 4.7	92.97	100.00	91.44	86.62	–	82.18	83.96	29.47
	DeepSeek V4 Pro	90.54	100.00	87.66	78.90	–	83.05	80.04	128.10
Normal	Optimization-Based + Human	88.72	100.00	89.07	72.71	–	87.05	77.74	5.68
	ChatGPT 5.5	86.28	100.00	93.82	66.27	–	77.64	73.19	39.18
	Gemini 3.1 Pro	88.23	100.00	94.10	71.50	–	86.68	71.72	41.53
	Claude Opus 4.7	89.18	100.00	96.09	68.24	–	84.54	78.12	48.32
	DeepSeek V4 Pro	87.63	100.00	91.36	71.74	–	83.95	75.13	111.94
Hard	Optimization-Based + Human	49.77	96.40	78.32	26.94	20.75	54.86	52.14	125.65
	ChatGPT 5.5	65.09	100.00	94.97	41.22	42.25	63.19	67.24	63.96
	Gemini 3.1 Pro	62.81	100.00	92.58	39.60	33.75	59.10	60.65	53.08
	Claude Opus 4.7	64.74	99.86	94.22	35.28	40.50	62.15	65.88	60.10
	DeepSeek V4 Pro	59.03	98.57	83.90	30.76	34.75	61.36	59.07	150.75
Expert	Optimization-Based + Human	21.53	78.27	42.13	1.93	0.00	42.07	39.87	339.85
	ChatGPT 5.5	40.15	93.24	67.01	13.23	2.67	60.15	61.25	74.51
	Gemini 3.1 Pro	38.28	90.09	61.30	12.02	0.00	58.72	55.82	66.20
	Claude Opus 4.7	39.81	92.43	65.55	9.13	0.00	58.99	60.11	78.29
	DeepSeek V4 Pro	37.55	90.95	61.09	3.54	0.00	53.43	54.43	174.28

In the simple and normal settings, which contain no runtime perturbations, the baseline scores 92.59 and 88.72, within the corresponding LLM ranges, while requiring only 3.42 and 5.68 s for planning. These times are 88.4% and 85.5% lower than those of the fastest LLM at the respective difficulty levels. In the hard and expert settings, the baseline scores fall to 49.77 and 21.53, which are 9.26 and 16.02 points below the weakest LLM, while planning time increases to 125.65 and 339.85 s. The hard value exceeds three of the four LLM configurations, and the expert value exceeds all four. The clear latency advantage observed in the fixed episodes therefore does not carry over to the episodes that require runtime replanning.

Figure 6 summarizes the total scores and planning times of all five configurations, while Figure 7 compares their mission-level profiles. Among the LLM configurations, Gemini 3.1 Pro and Claude Opus 4.7 perform best in the simple and normal settings, whereas ChatGPT 5.5 leads the hard and expert settings with scores of 65.09 and 40.15 and retains the most balanced profile. Averaged over all four levels, Claude Opus 4.7

obtains the highest LLM score of 71.67, followed by Gemini 3.1 Pro (70.64), ChatGPT 5.5 (70.53), and DeepSeek V4 Pro (68.68). DeepSeek V4 Pro records the lowest coordination and mission-time scores among the LLM configurations in both the hard and expert settings. The Optimization-Based + Human baseline averages 63.15, primarily because of its lower hard and expert scores. The changing ordering shows that no LLM configuration dominates across all levels of episode pressure.

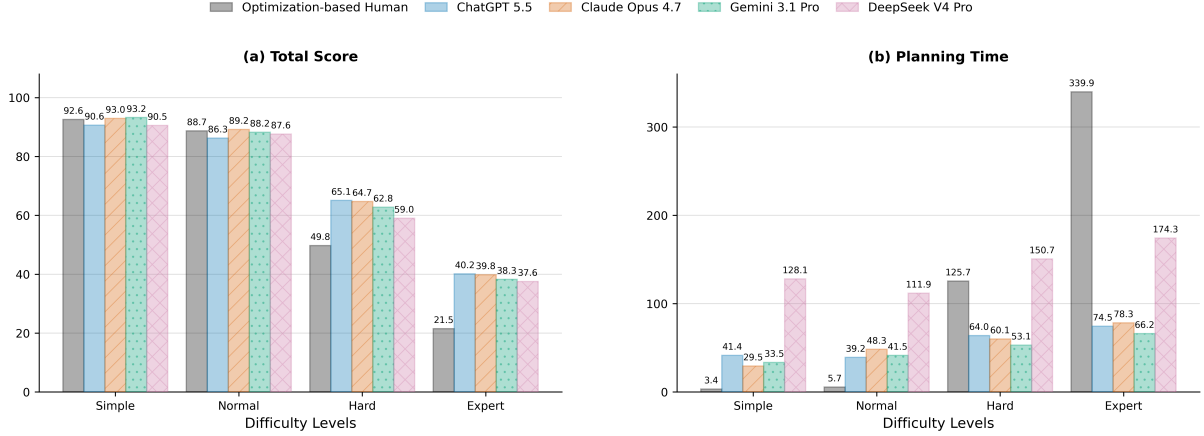


Figure 6: Total scores and planning times of the Optimization-Based + Human baseline and four LLM-based configurations across episode difficulty levels. Higher is better for total score, whereas lower is better for planning time.

Taken together, the results suggest a practical division between the two planning approaches under the evaluated conditions. Conventional optimization offers comparable mission quality with much lower latency in the fixed simple and normal episodes. In the hard and expert episodes, where greater operational pressure is coupled with natural-language runtime perturbations, the LLM configurations maintain higher mission scores and generally lower replanning latency than the human-assisted baseline. The comparison therefore does not establish the universal superiority of either approach; rather, it identifies adaptive event interpretation and plan revision as the setting in which LLM planning provides its clearest advantage.

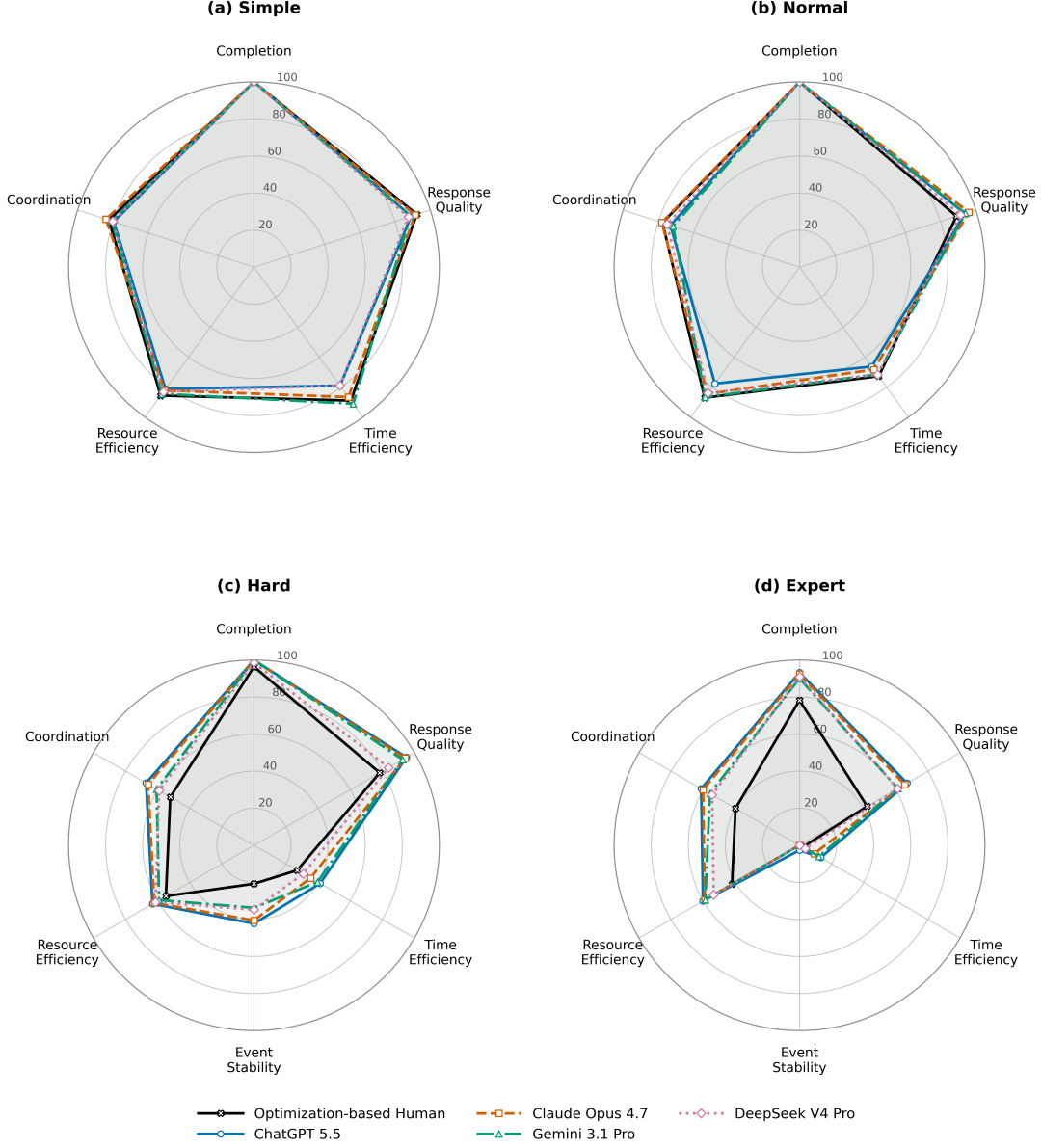


Figure 7: Mission-level performance profiles of the Optimization-Based + Human baseline and four LLM-based configurations at each difficulty level. Larger radial values indicate better performance. Event stability is omitted from the simple and normal settings because they contain no runtime perturbations.

5.3 Ablation Studies

Table 6 reports the ablation results with DeepSeek V4 Pro fixed as the foundation model. Removing Validation causes a modest but consistent reduction across all four difficulty levels, with drops of 2.15, 2.33, 3.80, and 2.06 points from simple to expert. The limited change in average mission score does not imply that the module remains inactive: the validation loop is triggered 5, 5, 4, and 6 times in the simple, normal, hard, and expert settings, respectively. These interventions suggest that Validation primarily serves as an occasional reliability safeguard against invalid or tactically implausible plans rather than as a continuous source of score improvement.

Figure 8 visualizes the score change of each ablated system relative to the full configuration.

Table 6: Ablation results with DeepSeek V4 Pro as the foundation model. Values report the mean overall score S over 20 episodes at each difficulty level. The best result in each column is shown in bold.

Configuration	Overall score S			
	Simple	Normal	Hard	Expert
Full	90.54	87.63	59.03	37.55
w/o Validation	88.38	85.30	55.22	35.50
w/o Memory	88.57	86.94	49.35	22.10
w/o Runtime Replanning	90.33	88.14	35.86	12.98

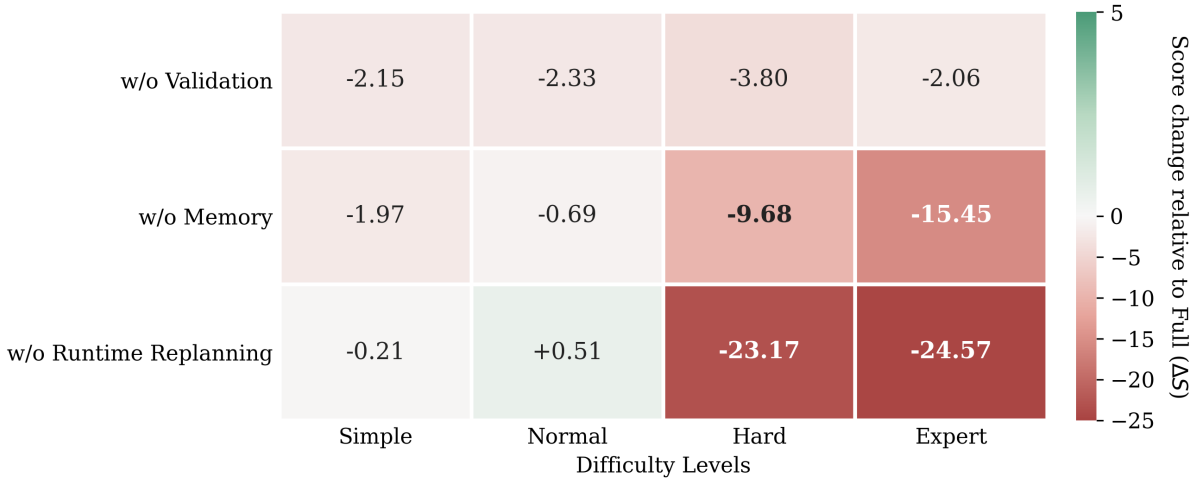


Figure 8: Change in overall score relative to the full system for each ablated variant and difficulty level. Negative values indicate performance degradation, while positive values indicate improvement.

The contribution of Memory becomes more pronounced as task demands increase. Removing both memory modules reduces the score by only 1.97 points in the simple setting and 0.69 points in the normal setting, but the reduction grows to 9.68 points in hard episodes and 15.45 points in expert episodes. This pattern indicates that retrieved cases and distilled lessons are most useful when resource constraints, event interactions, and scheduling pressure make planning less routine.

Disabling Runtime Replanning has little effect in the simple and normal settings, where no perturbations are triggered. In contrast, its score falls from 59.03 to 35.86 in hard episodes and from 37.55 to 12.98 in expert episodes. The sharp degradation confirms that an initially valid plan is insufficient once the execution state changes, and that runtime repair is the key mechanism for maintaining mission performance under perturbations.

5.4 Fine-Tuned Planner Evaluation

Table 7 compares the Gemini 3.1 Pro teacher planner with the original and fine-tuned Qwen3-8B planners. Fine-tuning substantially improves the compact model on both difficulty levels. The overall score increases from 63.77 to 90.87 in the simple setting and from 67.19 to 87.39 in the normal setting. Averaged over the two levels, fine-tuning yields an absolute gain of 23.65 points and raises Qwen3-8B to 98.24% of the Gemini

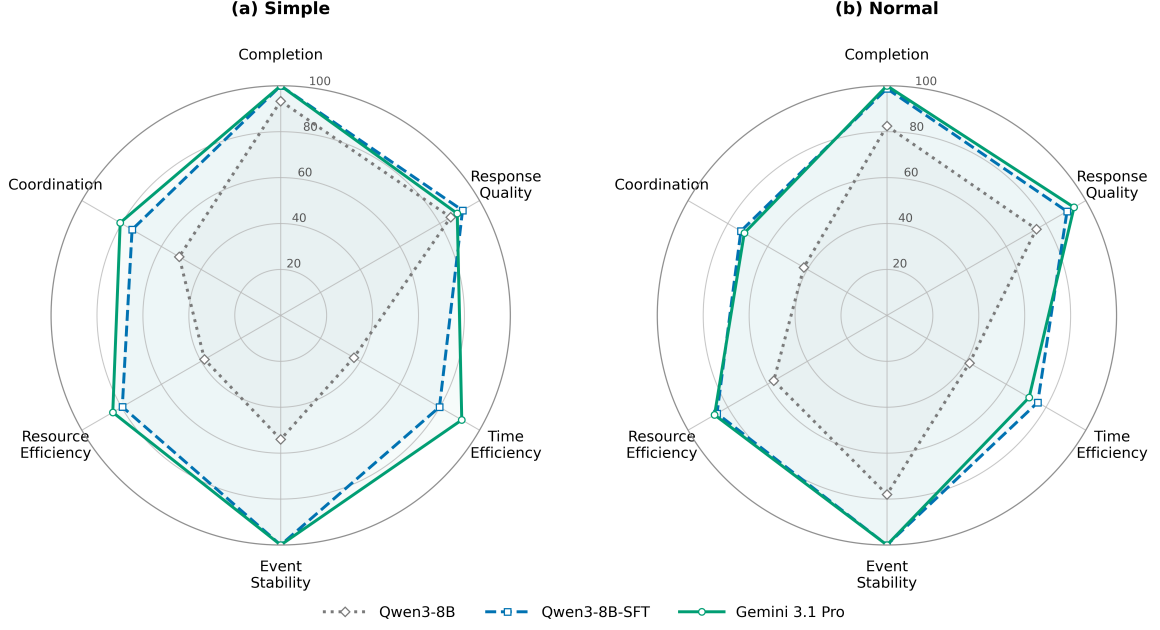


Figure 9: Mission-level profiles of the three planners in the simple and normal settings. Larger radial values indicate better performance. Event stability is not interpreted because neither setting contains runtime perturbations.

planner’s average score.

Table 7: Comparison of the Gemini 3.1 Pro teacher planner, the original Qwen3-8B planner, and the fine-tuned Qwen3-8B planner. Mission-level scores are normalized to $[0, 100]$ and higher is better. For planning time and token usage, lower is better. The best value at each difficulty level is shown in bold when it is unique.

Difficulty	Planner	S	Mission-level score					Planning overhead	
			Completion	Response	Time	Resource pressure	Coordination	Planning time (s)	Tokens
Simple	Gemini 3.1 Pro	93.23	100.00	88.78	91.06	84.45	80.71	33.47	11,730
	Qwen3-8B	63.77	93.34	85.48	36.77	38.33	51.07	18.37	3,804
	Qwen3-8B-SFT	90.87	100.00	91.44	79.83	79.66	74.73	12.95	3,485
Normal	Gemini 3.1 Pro	88.23	100.00	94.10	71.50	86.68	71.72	41.53	14,975
	Qwen3-8B	67.19	82.50	75.17	41.50	56.91	41.78	17.52	3,844
	Qwen3-8B-SFT	87.39	98.75	90.57	75.83	85.45	73.43	15.07	3,701

Figure 9 shows that the improvement extends across the mission-level performance profile. The original Qwen3-8B exhibits its largest deficits in mission time, resource pressure, and coordination, whereas fine-tuning substantially expands its profile toward that of Gemini 3.1 Pro. The fine-tuned planner restores near-complete task coverage and slightly exceeds Gemini 3.1 Pro in simple-level response and normal-level mission time and coordination. This indicates that the validated demonstrations transfer not only output-format compliance but also task-allocation and resource-management behavior to the smaller model.

The fine-tuned planner also reduces planning overhead under the evaluated deployment setting. Its mean planning time across the two levels is 14.01 s, compared with 37.50 s for Gemini 3.1 Pro, while its mean planning-token usage falls from 13,353 to 3,593. At the individual difficulty levels, this corresponds to planning-time reductions of 61.3%

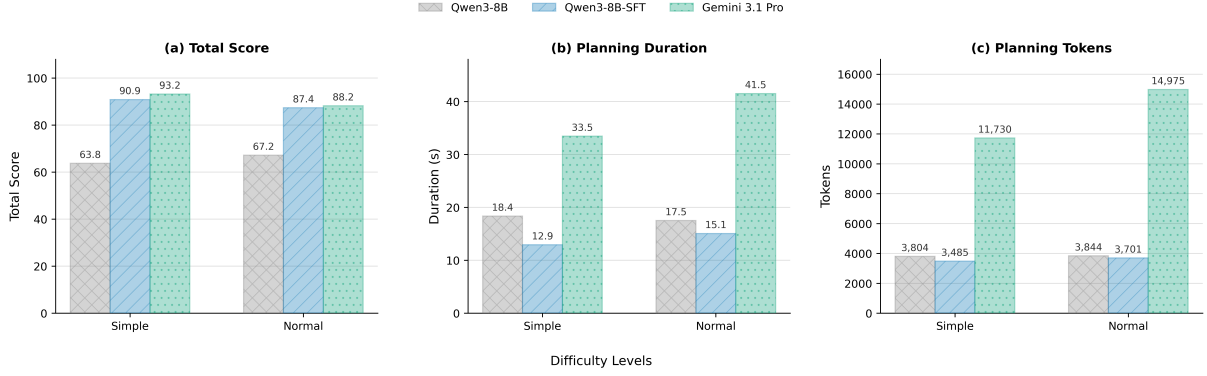


Figure 10: Score-efficiency comparison of the three planners: (a) total score, (b) planning duration, and (c) planning-token usage. Lower values are better in (b) and (c).

and 63.7%, and token reductions of 70.3% and 75.3%, for simple and normal episodes, respectively. Figure 10 summarizes this performance-efficiency trade-off: fine-tuning closes most of the score gap to the teacher while reducing both planning latency and token usage. These results show that domain specialization enables Qwen3-8B to retain most of the teacher planner’s mission performance while providing a more efficient locally deployable alternative for initial mission planning.

6 Conclusion

This paper formulates multi-UAV wildfire rescue as a closed-loop decision problem connecting aerial situation understanding, resource-constrained planning, and runtime adaptation. Based on this formulation, we develop a modular system that integrates Vision Analysis, Mission Planning, Validation, Episodic and Lesson Memory, and Runtime Replanning. We also construct a real-UAV-informed AirSim evaluation suite containing 80 fixed episodes across four difficulty levels and six complementary mission metrics.

Comparison with the Optimization-Based + Human baseline shows that conventional optimization matches LLM mission performance with substantially lower planning latency in simple and normal episodes, but loses its clear latency advantage when hard and expert episodes require runtime adaptation to natural-language perturbations. Among the four LLM configurations, Gemini 3.1 Pro and Claude Opus 4.7 perform strongly at lower difficulty, while ChatGPT 5.5 leads in hard and expert episodes. Ablation results show that Validation provides an occasional reliability safeguard, Memory becomes increasingly valuable as resource and scheduling pressure grows, and Runtime Replanning is critical under dynamic perturbations. Fine-tuning Qwen3-8B with validated planning demonstrations retains 98.24% of the Gemini 3.1 Pro planner’s average score on simple and normal episodes, while reducing mean planning time by approximately 63% and planning-token usage by 73%. These results support lower-cost local deployment of the initial planning module.

The current evidence remains limited to a calibrated simulation, a fixed episode set, and one real-platform configuration. Moreover, the specialized planner has not yet been evaluated under hard and expert runtime conditions. Future work will extend the evaluation to real-flight and mixed real-sim trials, more diverse platforms and fire dynamics, and larger fleets, while developing more selective memory retrieval and more robust, efficient replanning for high-pressure missions.